

MI 304 Research Paper

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Introduction

Social media is a large part of our society today, it allows us to make connections and communicate with people across the world. Social media is a valuable tool in our society but there are many different factors at play to make it run. Most social media platforms like Instagram, TikTok, YouTube, etc are powered by recommendation systems. Recommendation systems create the social media feeds of every single user across the platforms and the systems are powered by artificial intelligence. Many platforms like Instagram and TikTok have features like the “For You Page” or the “Explore Page” even though the content provided by these features isn’t made up of people users choose to follow, it is made up of content that the recommendation system recognizes as an interest of the user.

For example TikTok, uses a recommendation system called Monolith to power the “For You Page” feature. Monolith uses artificial intelligence to analyze users' interactions with content like watch time, likes, comments and to analyze the content itself like hashtags, captions, and sounds (TikTok Support, 2025). By analyzing the content this way the system is able recognize the type of content the user is interested in and what they like about it. After gathering this data, the system is able to send the users more content that is similar and increase the users TikTok usage. If the user is seeing content they enjoy they will continue to use the app.

The issue that arises with users constantly being exposed to content that they agree with is that echo chambers can form. Recommendation systems rank the likelihood that a user will interact with content based on buzz words, sounds, music, hashtags they have liked, commented on, or given significant watch time to in the past (Meserole, 2022). To gage this I thought back to the last time that I saw content on my For You Page or Explore Page that I genuinely disliked or had no interest in at all, I couldn’t remember the last time this happened. This is what happens

for the majority of people on social media especially people who are on social media for hours a day multiple times a day, the more content a user engages with the more data the system has to work with to recommend similar content. The study “Echo Chambers in Online Social Networks: A Systematic Literature Review” found that social networks themselves inherently contribute to the emergence of echo chambers (Mahmoudi et al., 2024). Echo chambers occur on social media when users are only exposed to content and opinions that they agree with. Echo chambers can lead to polarization, spread of misinformation, and hostility when confronted with differing opinions. When people only hear one side of an argument or opinion they become more extreme in their beliefs and experience confirmation bias which only leads to solidifying the opinion. Echo chambers can foster an “us vs them” mentality among its members, after only being exposed to opinions similar to their own users can become hostile and defensive when being confronted with the latter. Echo chambers can also contribute to misinformation spreading online. People that are a part of echo chambers are more susceptible to misinformation because they are unlikely to seek out information that may differ from what they believe in. Echo chambers throw off the user's perception of society, when individuals feel their opinions are widely accepted they are less likely to question their perspectives (Koehler, 2024). All of these aspects of echo chambers can affect the mental health of social media users and influence their perception of reality.

Methods

In the last decade we have seen a major switch to what content is shown on social media. Social media went from a place to share photos and videos to a place for political discourse, trolling, and comparison. I conducted a small study among my peers, all of which are on social media a fair amount. When asked if social media has the power to make them feel anxious,

overwhelmed, etc 83% of respondents said yes to some degree, ranging from sometimes to very often. At this point every person on social media has been exposed to some type of toxicity online and political discourse plays a major part in that toxicity. In the study I conducted, 50% of respondents stated that the toxic content they see online is most often relating to politics and misinformation. How do AI powered recommendation systems play a part in creating echo chambers that foster this toxicity online?

In general, on social media the continuous exposure to similar content and opinions can shape the perceptions of users. The article “AI-Driven Content Filtering on Instagram and Its Impact on Youth Lifestyle and Interaction” finds that Instagram’s algorithm actively shapes the attitudes and behaviors of adolescents through continuous exposure to homogeneous content (Misnawati et al, 2025). This study was conducted using qualitative methods, there were 30-40 minute interviews held with each participant. After the interview thematic analysis was conducted to find the main theme that each participant touched on in their answers. The dominant themes that were identified were internalization of ideal beauty standards, algorithmic reinforcement of lifestyle conformity, and homogeneity in social interactions (Misnawati et al, 2025) . So, all in all the study found that repeated algorithmically curated content does shape the perception of teens regarding beauty, fashion, and lifestyle. We can only assume that this theme of algorithmic reinforcement would also translate into political discourse online.

In terms of political discourse the article “An Extensive Examination of Toxicity, Polarization, and Biasing in Political Conversation on Social Media” explored how algorithms and echo chambers contribute to the creation of biased information environments, potentially altering political discourse and affecting user perceptions (Garg et al., 2024). The study found that the polarization of political discourse online can both energize supporters and demobilize

opponents (Garg et al., 2024). In terms of political discourse, being presented with supporting information, true or not, will encourage supporters and those who are a part of echo chambers will be unlikely to fact check this information. This decision to not fact check information is a common fault among people who participate in political discourse online and contributes to the toxic environment. At a point echo chambers can progress from being a bubble of people with the same opinion to a bubble of people spreading misinformation, knowingly or unknowingly, in support of their cause. If on the off chance someone with a differing opinion is presented with this information it can prompt them to do additional research to fact check but, one user trying to relay information that doesn't agree with the consensus of the echo chamber will most likely not result in the echo chamber accepting the information as truth. This rejection will lead to the "opponent" no longer trying to share information that corrects the misinformation being spread.

As mentioned above, I conducted a small survey among my peers to get an idea of how the average 18-21 year old feels about the toxic environments online and how aware they are of how recommendation systems influence their social media feeds and involvement in echo chambers. All of the participants stated that they use social media platforms that rely on recommendation systems multiple times a day. All participants recognize that they often find themselves in echo chambers online. 83% of participants rely on social media as their primary source of news or political information. Lastly, 83% of participants reported that social media makes them feel anxious, overwhelmed, etc ranging from sometimes to very often. Conducting this study even though it was small, provided insight on how people who use social media daily are impacted by echo chambers and how they are affected by toxic environments online.

Findings

The study “Recommendation System as a Social Determinant of Health” found that recommendation algorithms are affecting individuals and society as a whole by influencing social behavior, polarization, and mental health of users (Goswami, 2024). It is a known phenomenon that consistent and long term social media/internet use can affect the mental health of young people. This has been observed in many instances throughout the past 15 years and has only increased as social media has become more popular. As social media has progressed it has morphed into something completely different than what it was in the beginning and it has changed the perceptions of all users in one way or another. This change in perception is only being exacerbated by algorithmically curated content. Personalized social media feeds have allowed echo chambers to form across all aspects of social media, including political discussions and simple lifestyle videos.

Discussion

Based on the information I have gathered from my review literature on the topics of echo chambers, recommendation systems, and mental health and the study I conducted on my own I find it extremely important to recognize the harm that is being done to users. At such a heightened time of political discourse and misinformation, I think it is in the best interest of everyone on social media to become aware of the role that echo chambers may play in influencing their opinions and making them more susceptible to misinformation. I also believe that the responsibility to protect users falls on these social media platforms who rely on recommendation systems to power their apps. Echo chambers created by algorithmically curated content have the power to influence mental health and public perceptions. Therefore, social media networks need to alter the recommendation algorithms that are currently in place to expose users to more diverse content to protect their mental health.

Limitations

The main limitation I found while conducting this research was the literature and studies that were available did not completely encapsulate my research question. I was able to find studies that explored the individual pieces of my research, but I was not able to find a reliable study that included all aspects of my question. This is why I chose to conduct a survey of my own so I could gather information that touched on every aspect. In the future it would be interesting to see a study conducted that worked to find a relationship between recommendation systems creating echo chambers that included the perception and mental health of users around political discourse specifically.

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